

Discrete Structures

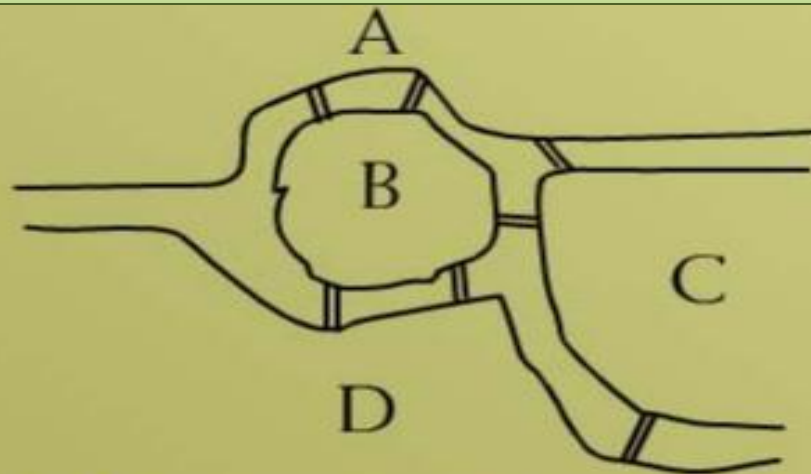
Lecture # 13

Dr. Muhammad Ahmad

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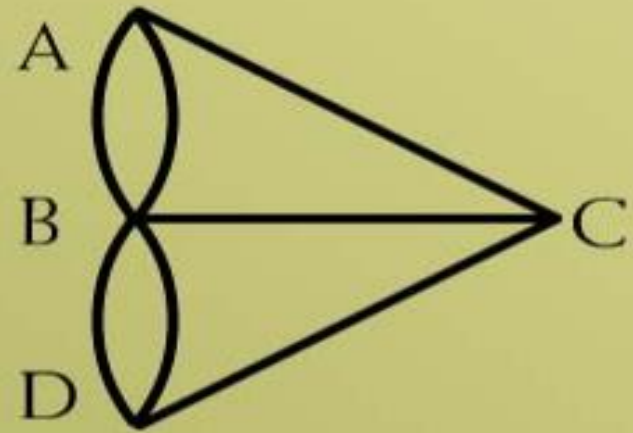
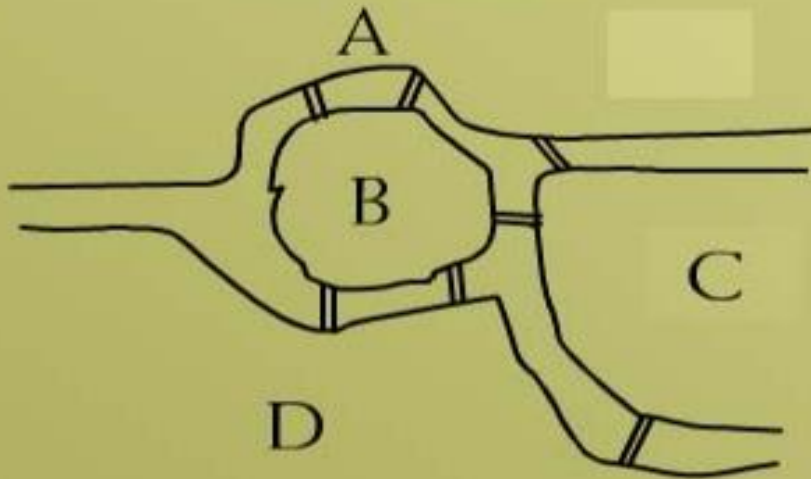
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KONIGSBERG BRIDGES PROBLEM



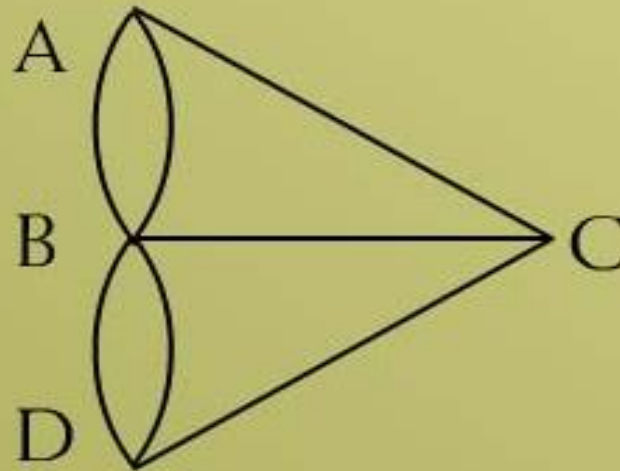
Is it possible for **a person** to take a walk around town, **starting and ending** at the **same location** and crossing each of the **seven bridges** exactly once?

SOLUTION



Is it possible to find **a route** through the **graph** that **starts and ends** at some **vertex A, B, C** or **D** and **traverses** each edge exactly once?

EQUIVALENT FORM OF BRIDGE PROBLEM



Is it possible to **trace** this **graph**, **starting** and **ending** at the **same point**, without ever **lifting** your **pencil** from the **paper**?

TERMINOLOGY

Let G be a **graph** and let v and w be **vertices** in graph G .

1. WALK

A **walk** from v to w is a **finite alternating sequence** of **adjacent vertices** and **edges** of G .

TERMINOLOGY

Thus a **walk** has the form

$$v_0 e_1 v_1 e_2 \dots v_{n-1} e_n v_n$$

where the **v's** represent vertices, the **e's** represent edges $v_0 = v, v_n = w$, and for all $i = 1, 2 \dots n$, v_{i-1} and v_i are **endpoints** of e_i .

The **trivial walk** from **v to v** consists of the **single vertex v**.

TERMINOLOGY

2. CLOSED WALK

A **closed walk** is a **walk** that **starts** and **ends** at the **same vertex**.

3. CIRCUIT

A **circuit** is a **closed walk** that does not contain a **repeated edge**.

Thus a **circuit** is a **walk** of the form

$$v_0 e_1 v_1 e_2 \dots v_{n-1} e_n v_n$$

where $v_0 = v_n$ and all the e_i 's are **distinct**

TERMINOLOGY

4. SIMPLE CIRCUIT

A **simple circuit** is a **circuit** that does not have **any other repeated vertex** except the **first and last**.

Thus a **simple circuit** is a **walk** of the form

$$v_0 e_1 v_1 e_2 \dots v_{n-1} e_n v_n$$

where all the **e_i 's** are **distinct** and all the **v_j 's** are **distinct** except that **$v_0 = v_n$**

TERMINOLOGY

5. PATH

A **path** from **v** to **w** is a **walk** from **v** to **w** that does not contain a **repeated edge**.

Thus a path from **v** to **w** is a **walk** of the form

$$v = v_0 e_1 v_1 e_2 \dots v_{n-1} e_n v_n = w$$

where all the **e_i**'s are **distinct** (that is **e_i ≠ e_k** for any **i ≠ k**).

TERMINOLOGY

6. SIMPLE PATH

A **simple path** from **v** to **w** is a path that does not contain a **repeated vertex**.

Thus a **simple path** is a **walk** of the form

$$v = v_0 e_1 v_1 e_2 \dots v_{n-1} e_n v_n = w$$

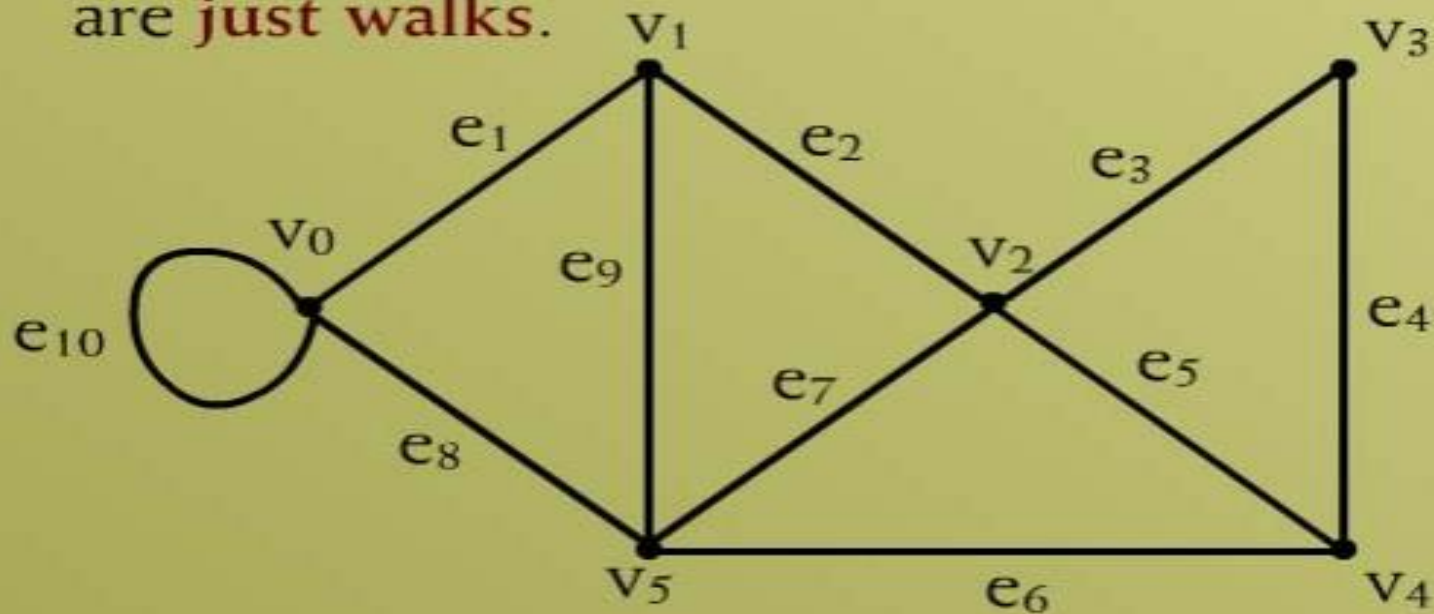
where all the **e_i's** are **distinct** and all the **v_j's** are also **distinct** (that is, $v_j \neq v_m$ for any $j \neq m$).

SUMMARY

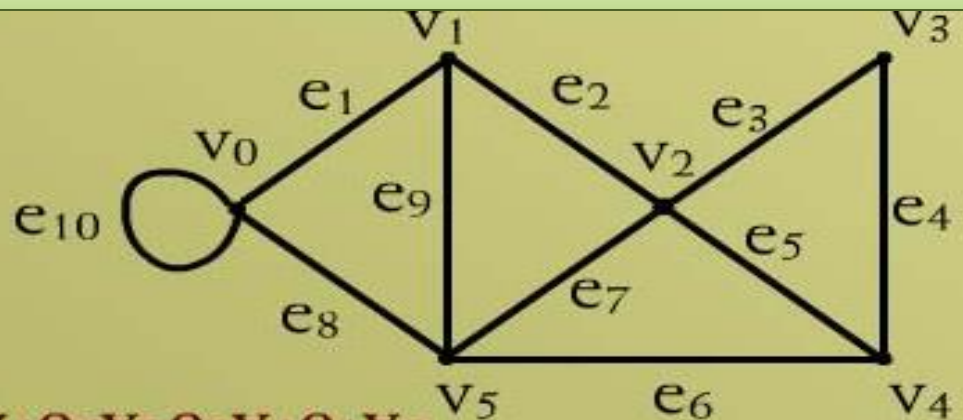
Criteria Terms	Repeated Edge	Repeated Vertex	Starts and Ends at Same Point
walk	allowed	allowed	allowed
closed walk	allowed	allowed	yes
circuit	no	allowed	yes
simple circuit	no	first and last only	yes
path	no	allowed	no
simple path	no	no	no

PROBLEM

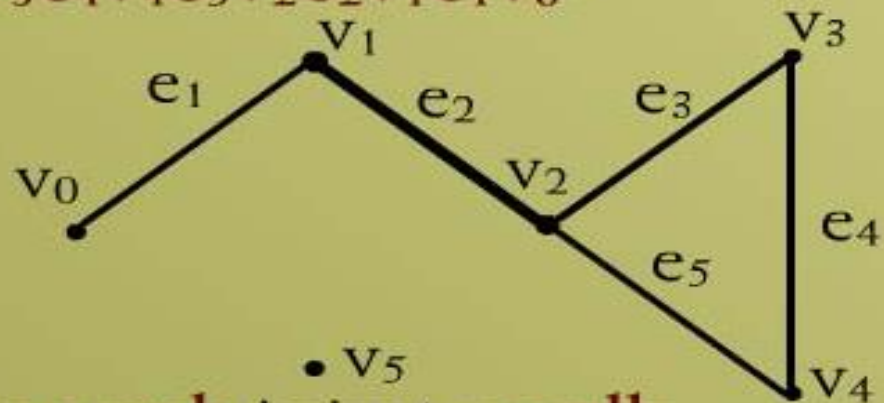
In the **graph** below, determine whether the following **walks** are **paths**, **simple paths**, **closed walks**, **circuits**, **simple circuits**, or are **just walks**.



SOLUTION

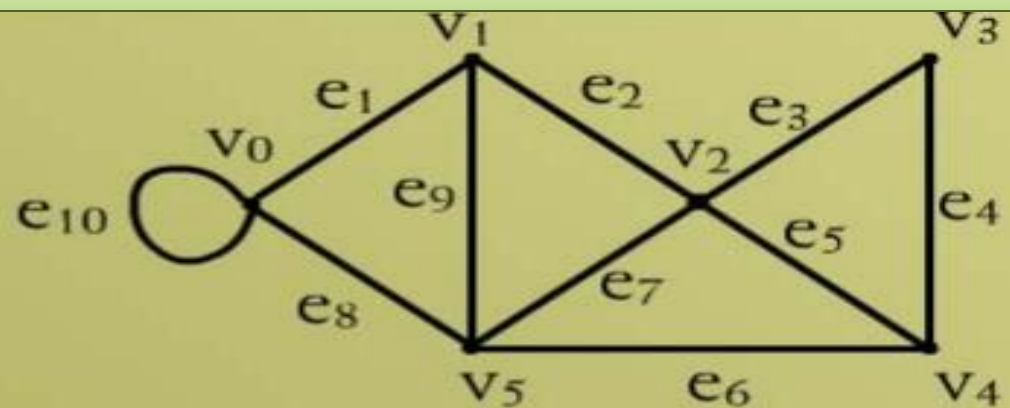


(a) $v_1 e_2 v_2 e_3 v_3 e_4 v_4 e_5 v_2 e_2 v_1 e_1 v_0$

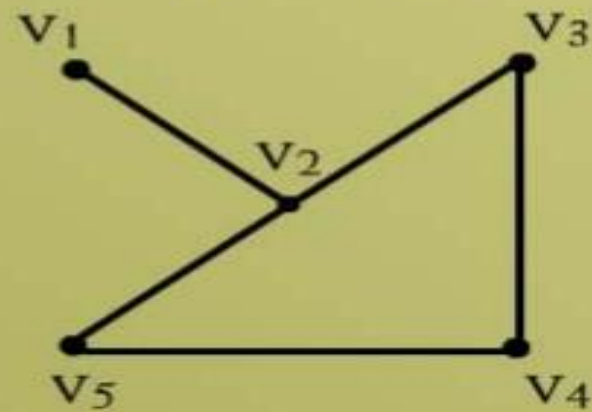


The **graph** is just a **walk**.

SOLUTION

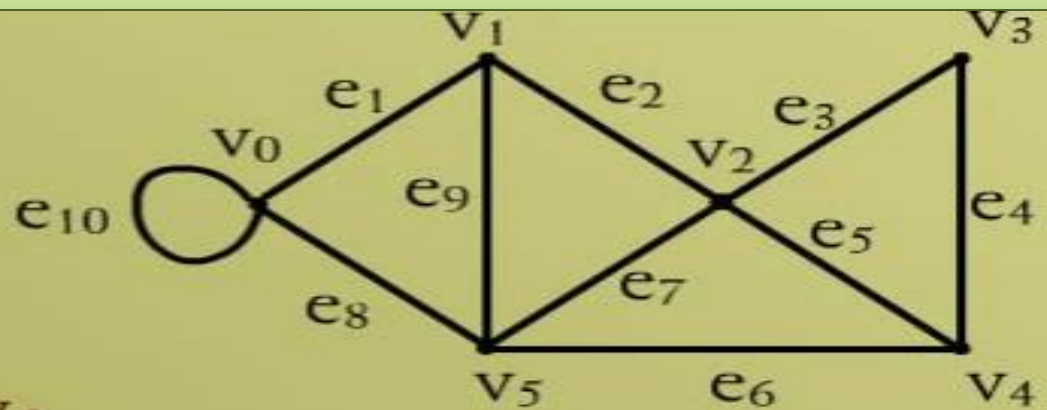


(b) $v_1v_2v_3v_4v_5v_2$

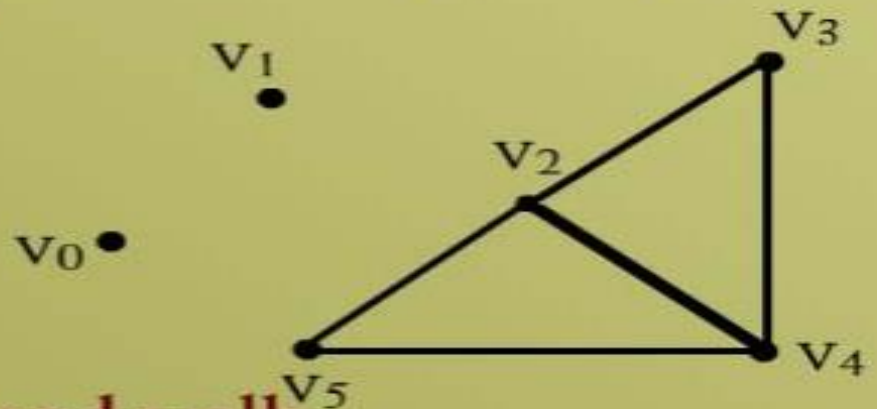


The graph is a path.

SOLUTION

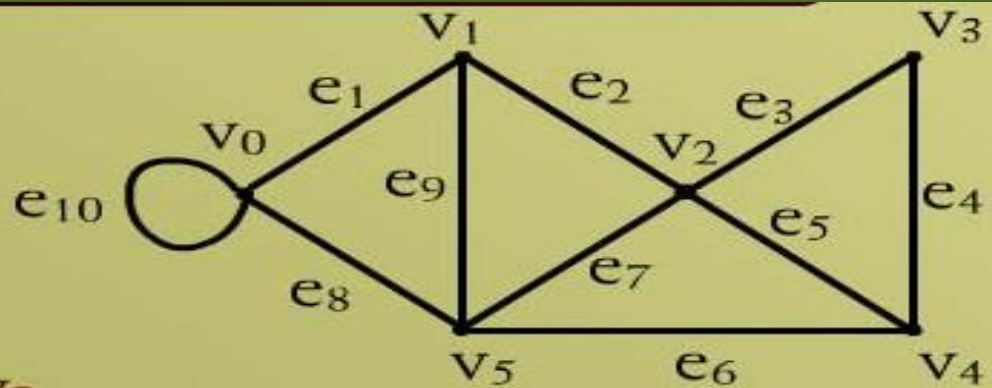


(c) $v_4v_2v_3v_4v_5v_2v_4$



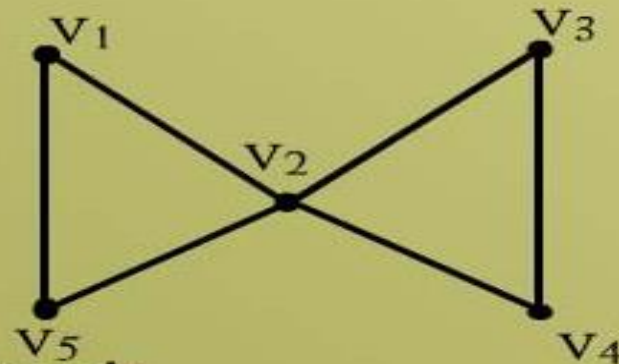
The **graph** is a **closed walk**.

SOLUTION



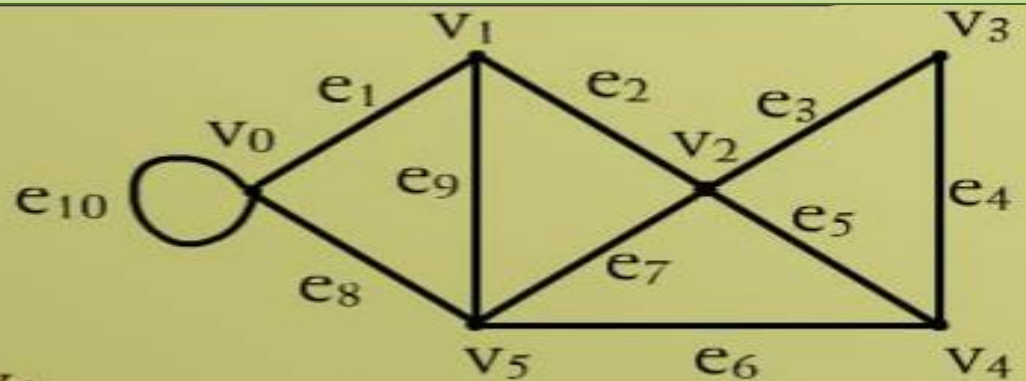
(d) $v_2v_1v_5v_2v_3v_4v_2$

$v_0 \bullet$

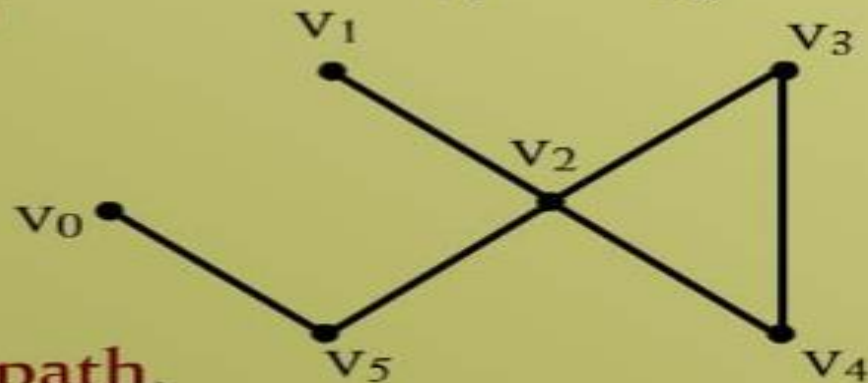


The graph is a circuit.

SOLUTION

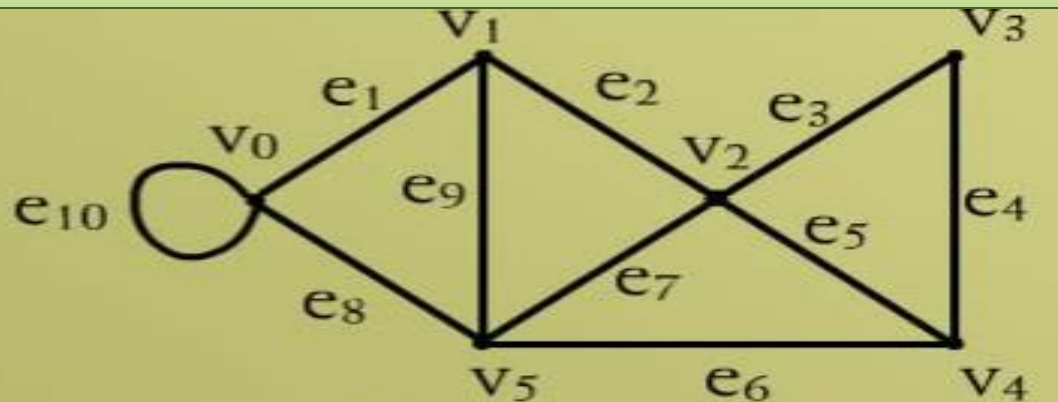


(e) $v_0v_5v_2v_3v_4v_2v_1$

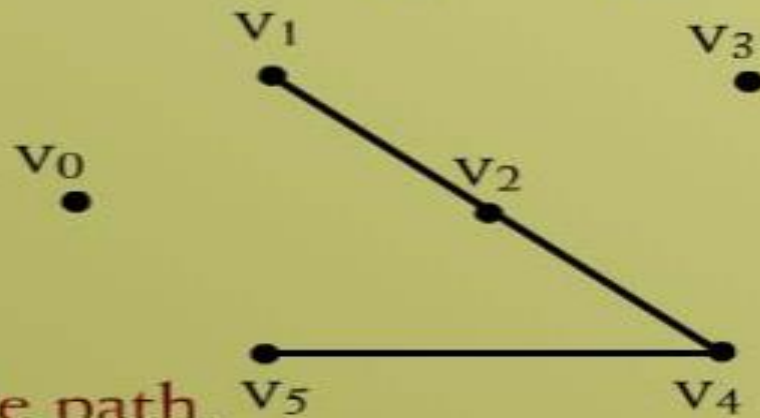


The **graph** is a **path**.

SOLUTION



(f) $v_5v_4v_2v_1$



The graph is a simple path.

CONNECTEDNESS

Let G be a **graph**. Two vertices v and w of G are connected **if, and only if**, there is a **walk** from v to w .

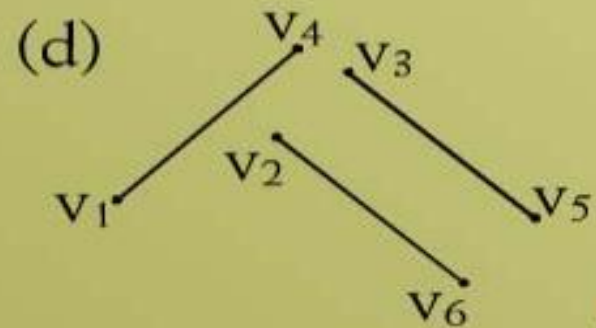
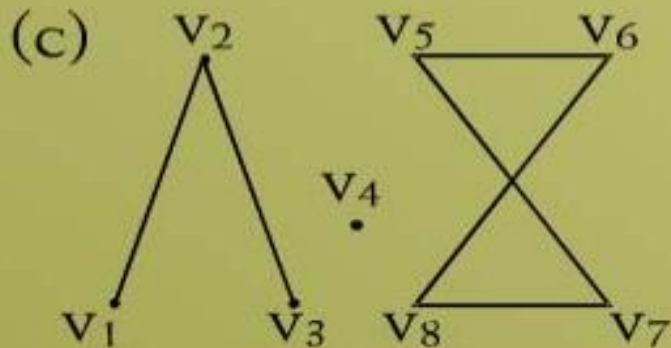
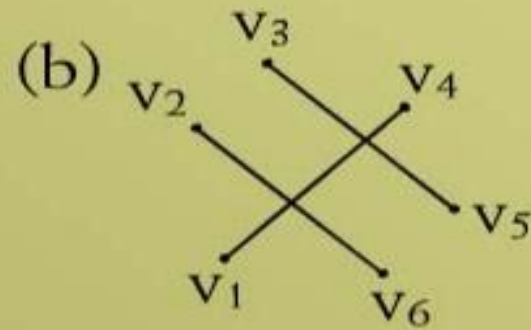
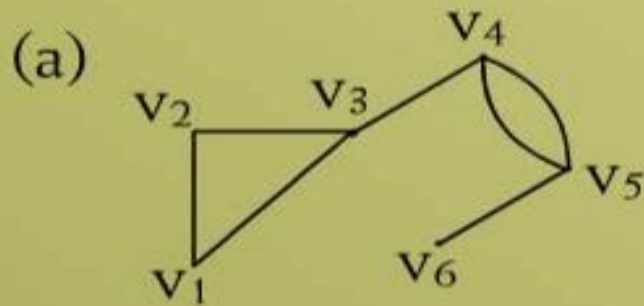
The **graph** G is connected **if, and only if**, given any **two vertices** v and w in G , there is a **walk** from v to w .

Symbolically:

G is **connected** $\Leftrightarrow \forall$ vertices $v, w \in V(G)$,
 $\exists$ a walk from v to w :

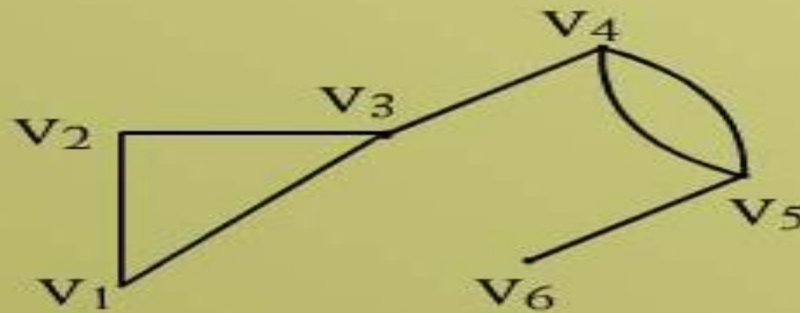
EXAMPLE

Which of the following **graphs** are **connected**?



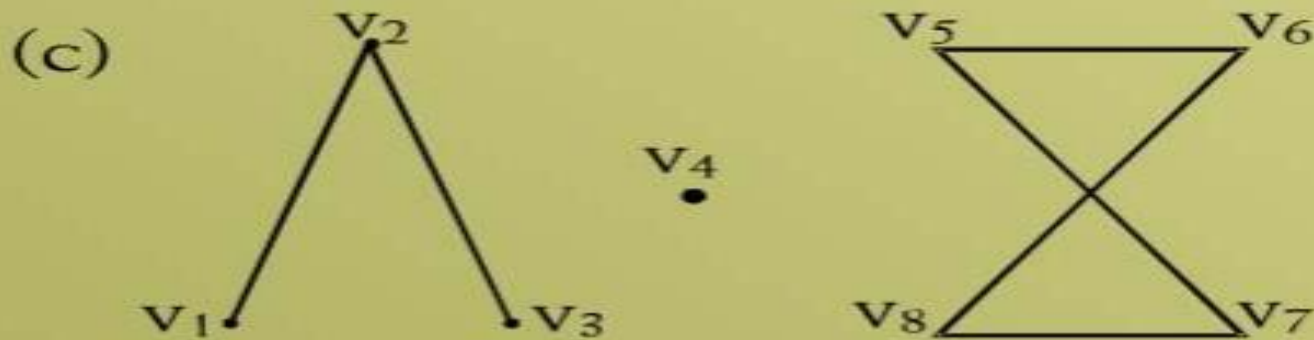
EXAMPLE

(a)



It has **six vertices** $v_1, v_2, \dots, v_6$.
The graph is **connected**.

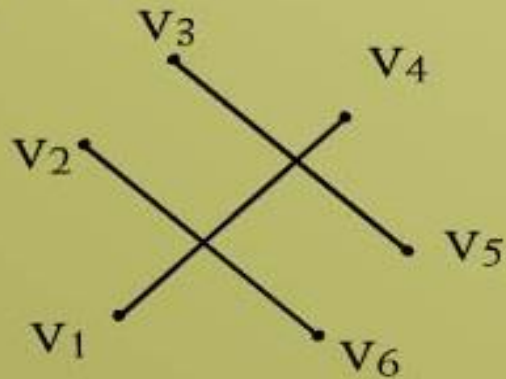
EXAMPLE



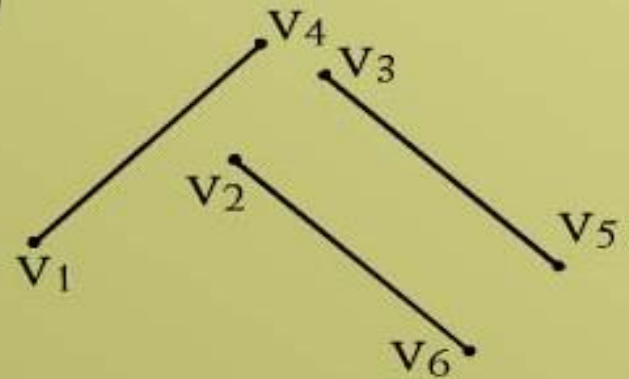
This **graph** is not **connected**.

EXAMPLE

(b)



(d)



So **graph (b) and (d) are not connected.**

EULER CIRCUITS

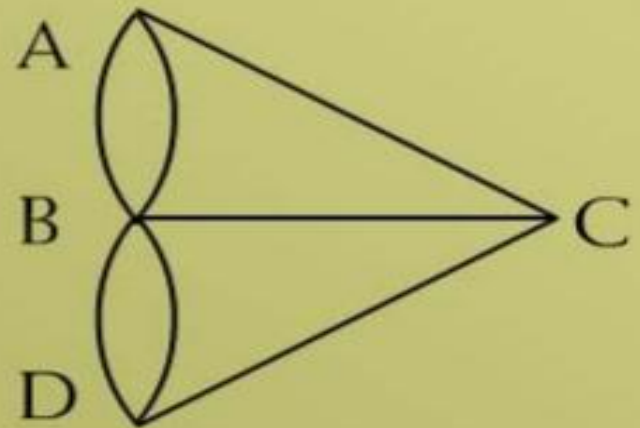
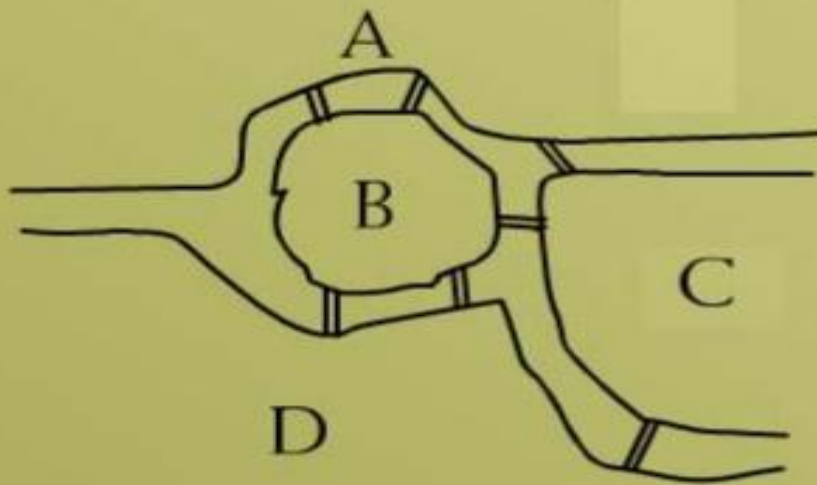
Let G be a **graph**. An **Euler circuit** of G is a circuit that contains every **vertex** and every **edge** of G .

That is, an **Euler circuit** of G is **sequence** of **adjacent vertices** and **edges** in G that starts and **ends** at the **same vertex**, uses every vertex of G at least once, and uses **every edge** of G exactly once.

EULER RESULT

A graph G has an Euler circuit if, and only if, G is connected and every vertex of G has an even degree.

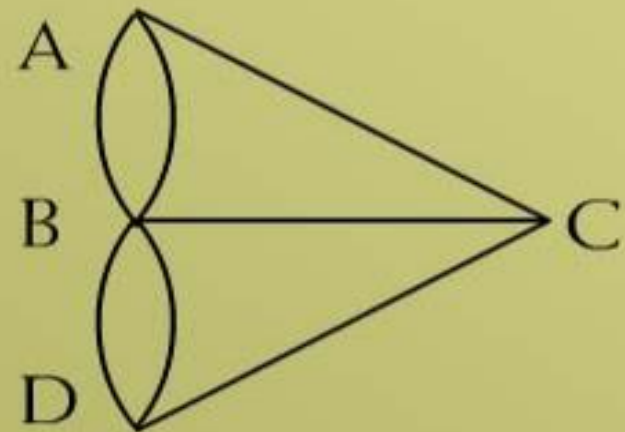
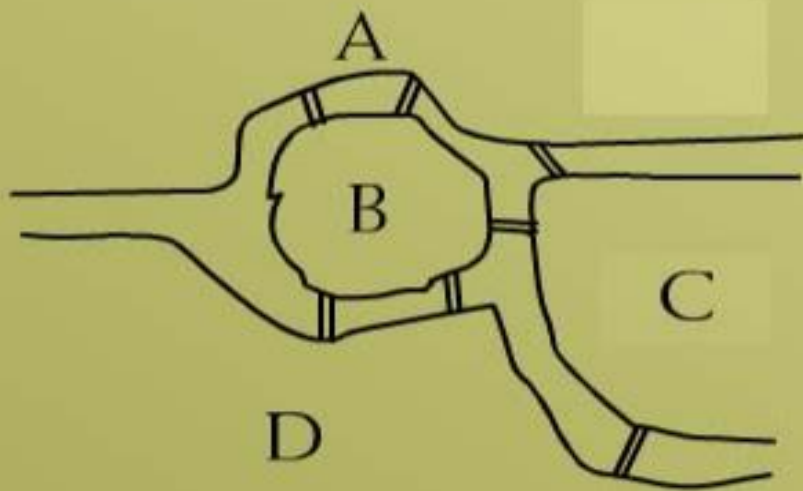
KONIGSBERG BRIDGES PROBLEM



$$\deg(a) = 3 = \deg(c)$$

$$\deg(b) = 5, \deg(d) = 3$$

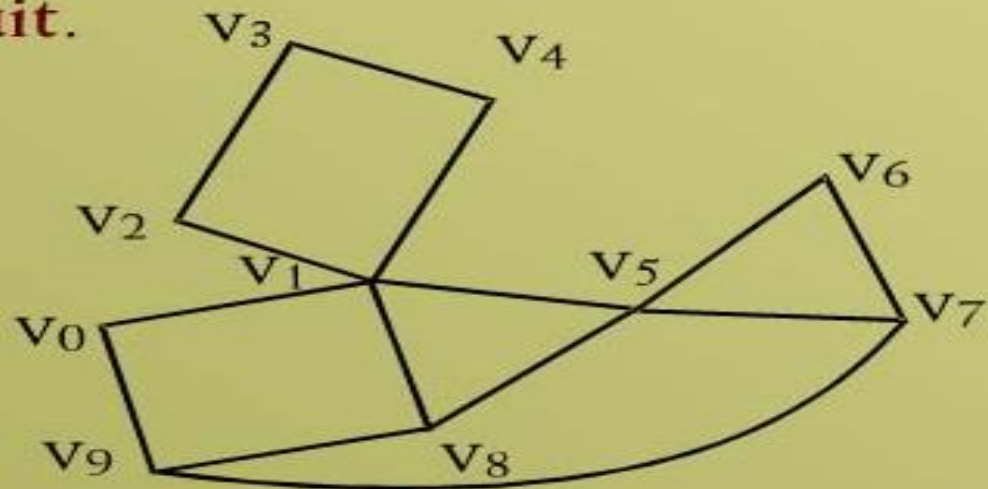
KONIGSBERG BRIDGES PROBLEM



No **vertex** has **even degree** so there is no possibility of an **Euler circuit**.

EXERCISE

Determine whether the following **graph** has an **Euler circuit**.

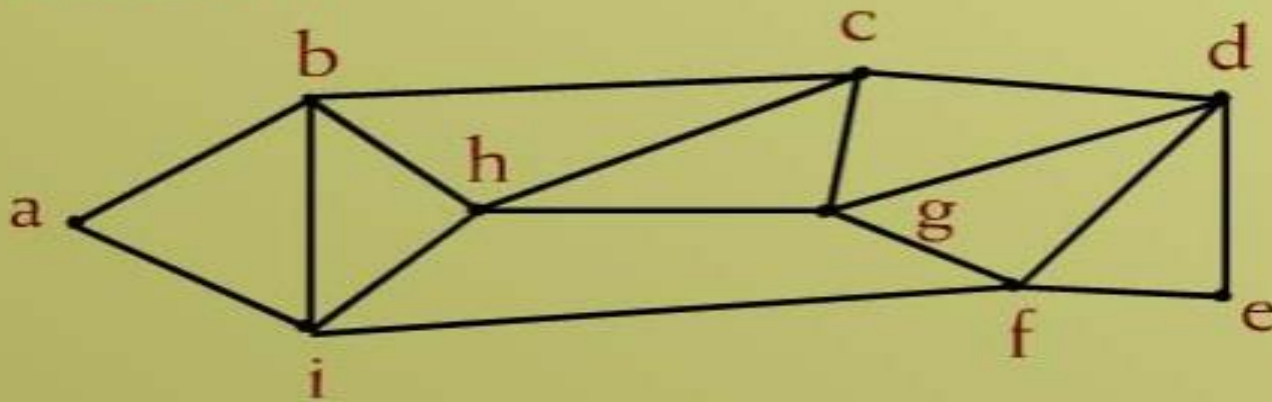


$\deg(v_3) = 2 = \deg(v_4) = \deg(v_2)$, $\deg(v_1) = 5$

As v_1 has **odd degree** so this **graph** can't have an **Euler circuit**.

EXERCISE

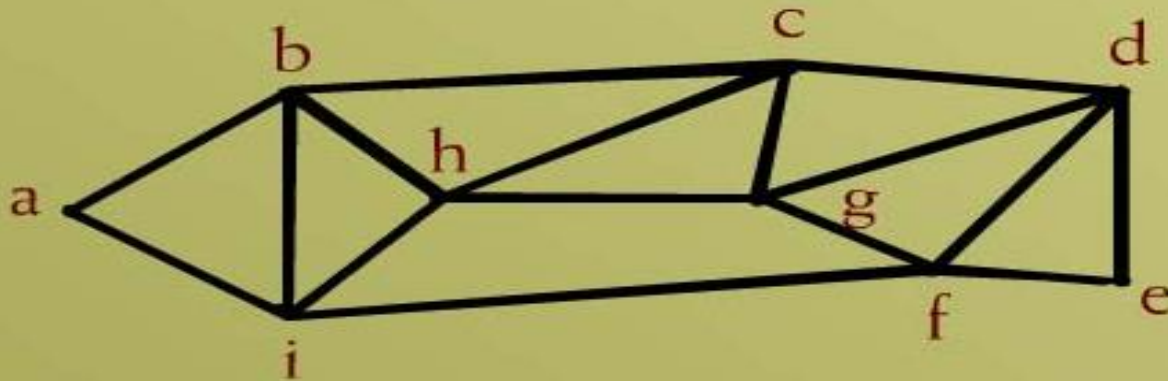
Determine whether the following **graph** has **Euler circuit**.



$$\begin{aligned} \deg(a) &= 2, \deg(b) = 4, \deg(c) = 4, \deg(d) = 4, \\ \deg(e) &= 2, \deg(f) = 4, \deg(g) = 4, \deg(h) = 4, \\ \deg(i) &= 4 \end{aligned}$$

EXERCISE

So the **every vertex** is of **even degree**, clearly **Euler theorem** is applicable. We should be able to find **Euler circuit** here:



Euler circuit: {a, b, c, d, f, e, d, g, f, i, h, g, c, h, b, i, a}.

EULER PATH

Let G be a **graph** and let v and w be two vertices of G .

An **Euler path** from v to w is a sequence of **adjacent edges** and **vertices** that starts at v , **end** at w , passes through every **vertex** of G at least once, and **traverses** every **edge** of G exactly once.

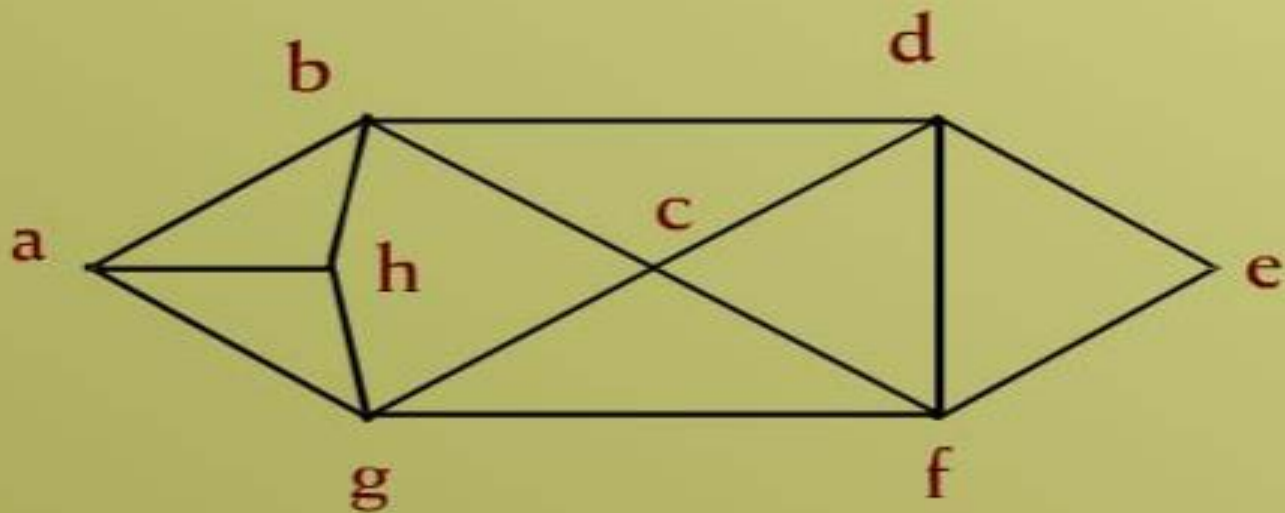
HAMILTONIAN CIRCUITS

Given a graph G , a **Hamiltonian circuit** for G is a **simple circuit** that includes **every vertex** of G .

That is, a **Hamiltonian circuit** for G is a **sequence** of **adjacent vertices** and **distinct edges** in which **every vertex** of G appears exactly once.

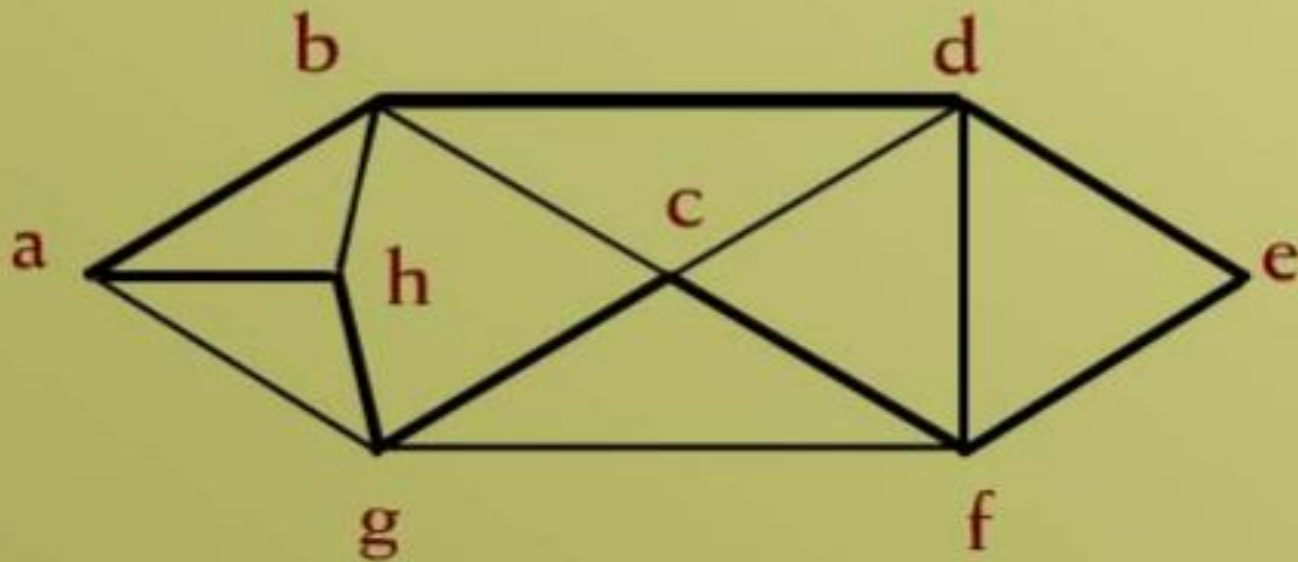
EXERCISE

Find **Hamiltonian Circuit** for the following graph.



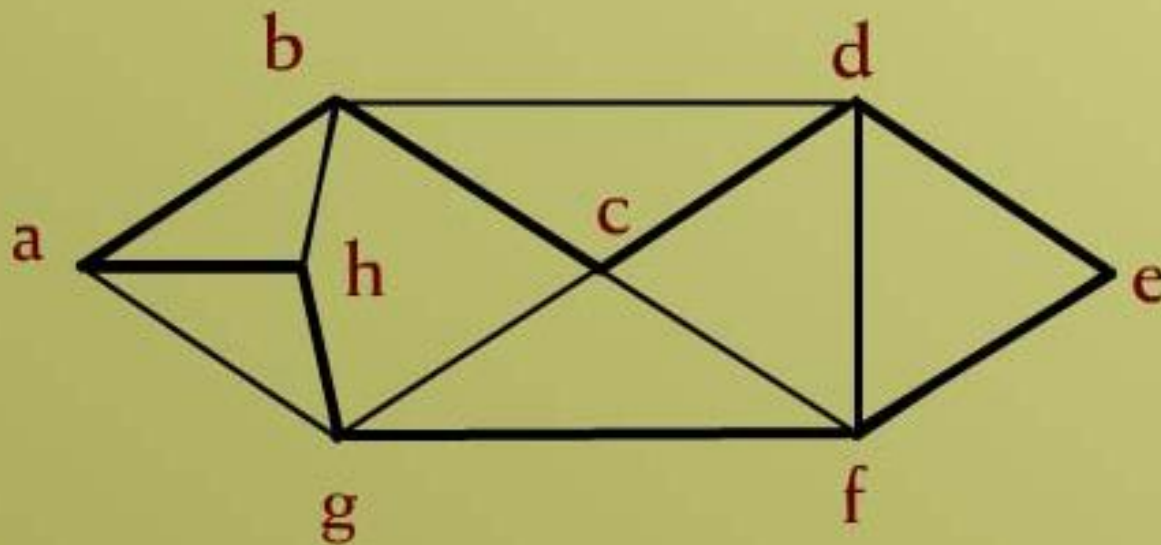
SOLUTION

The **Hamiltonian Circuit** is:



SOLUTION

Another **Hamiltonian Circuit** could be:



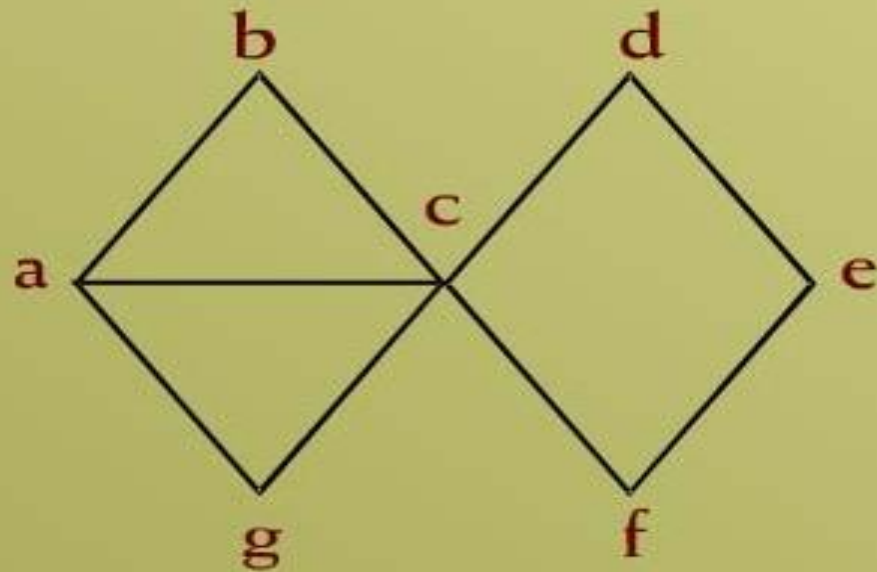
PROPERTIES

If a graph G has a **Hamiltonian circuit** then G has a **sub-graph** H with the following properties:

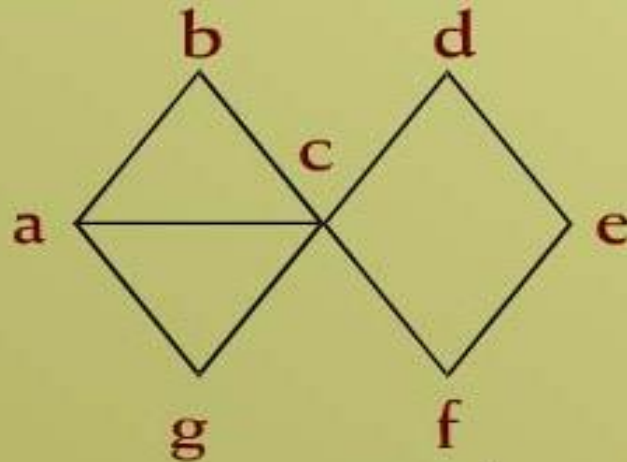
1. H contains every **vertex** of G .
2. H is **connected**.
3. H has the same number of edges as vertices.
4. Every vertex of H has **degree 2**.

EXAMPLE

Show that whether the **Hamiltonian circuit** is possible or not?



EXAMPLE



$\deg(c) = 5$, if we remove 3 edges from vertex c
then $\deg(a) < 2, \deg(b) < 2, \deg(g) < 2, \deg(d) < 2$

It means that this **graph** does not have a
subgraph with the desired properties, so the
Hamiltonian circuit is not possible.

Is the following graph a Hamiltonian graph? Give the explicit reason.

